

Assignment 3 - ELG 5218 Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Each item (a, b, ...) in the questions is worth 1 point unless otherwise noted. The total score is 37 points. You need 32 points to achieve 100.

Question 1: Multimodal Learning

Problem 1: From Mahalanobis to Cosine — Unifying Association Metrics

In classical tracking the association cost between a track with predicted measurement $\mathbf{H}\hat{\mathbf{z}}$ and innovation covariance $\mathbf{S}$ and a new measurement $\mathbf{y}$ is the Mahalanobis distance:

$$D_M^2 = (\mathbf{y} - \mathbf{H}\hat{\mathbf{z}})^\top \mathbf{S}^{-1} (\mathbf{y} - \mathbf{H}\hat{\mathbf{z}}).$$

In learned tracking the cost is derived from cosine similarity in embedding space:

$$C_{\text{emb}} = 1 - \cos(\mathbf{e}_{\text{track}}, \mathbf{e}_{\text{det}}).$$

- (a) Show that if the embedding encoder is a linear map $\mathbf{e} = \mathbf{W}\mathbf{x}$ applied to the physical state, and the embeddings are *not* normalized, then the squared Euclidean distance $\|\mathbf{e}_1 - \mathbf{e}_2\|^2$ in embedding space can be written as a Mahalanobis-like distance in the original state space. Identify the effective “covariance” matrix in terms of $\mathbf{W}$.
- (b) Explain precisely what additional expressive power is gained when f_θ is a deep nonlinear encoder instead of the linear map in (a). Give two concrete examples from multimodal tracking where this matters.
- (c) **Hybrid association cost (2 points).**

In multi-target tracking, the *data association* step must decide which new detections correspond to which existing tracks before a state update can be performed. The association step receives two lists:

- **Tracks:** each track i carries a predicted measurement $\mathbf{H}\hat{\mathbf{z}}_i$ with innovation covariance $\mathbf{S}_i = \mathbf{H}\mathbf{P}_{i|t-1}\mathbf{H}^\top + \mathbf{R}$, and a track appearance embedding $\mathbf{e}_i$.
- **Detections:** each detection j carries a measurement $\mathbf{y}_j$ and an appearance embedding $\mathbf{e}_j$.

A practitioner proposes the following **combined association cost**:

$$C_{ij} = \alpha \frac{D_M^2(i, j)}{\chi_{\text{gate}}^2} + \beta (1 - \cos(\mathbf{e}_i, \mathbf{e}_j)),$$

where

- $D_M^2(i, j) = (\mathbf{y}_j - \mathbf{H}\hat{\mathbf{z}}_i)^\top \mathbf{S}_i^{-1}(\mathbf{y}_j - \mathbf{H}\hat{\mathbf{z}}_i)$ is the squared Mahalanobis distance,
- χ_{gate}^2 is a *gating threshold* chosen from the χ^2 distribution with degrees of freedom equal to the measurement dimension n_y at a desired confidence level (e.g. $\chi_{0.99, n_y}^2$); dividing by it normalizes the Mahalanobis term so that $D_M^2/\chi_{\text{gate}}^2 \approx 1$ at the gate boundary,
- $\cos(\mathbf{e}_i, \mathbf{e}_j)$ is the cosine similarity between the track and detection embeddings,
- $\alpha, \beta \geq 0$ are weighting coefficients with $\alpha + \beta = 1$.

Answer the following:

- Identify precisely *where* in the tracking pipeline this cost is used. Draw or describe the sequence of steps in a single tracking cycle (predict $\rightarrow$ gate $\rightarrow$ associate $\rightarrow$ update) and indicate at which step C_{ij} appears.
 - Explain *why* a single metric is insufficient. Give one concrete scenario where the Mahalanobis term alone ($\beta = 0$) produces an ambiguous or incorrect association, and one scenario where the cosine term alone ($\alpha = 0$) fails.
 - Explain the role of dividing by χ_{gate}^2 . What would go wrong if the raw D_M^2 were added directly to the cosine term without normalization?
 - Discuss how α and β should be adapted depending on the operational scenario. Give one autonomous-driving example favoring α and one surveillance example favoring β .
- (d) In JPDA, association probabilities are computed as

$$\beta_{ij} \propto p(\mathbf{y}_j \mid \text{track } i) \propto \exp\left(-\frac{1}{2} C_{ij}\right),$$

where C_{ij} is the combined cost defined in part (c).

- Justify why the exponential-of-negative-cost form is a natural choice by identifying the distributional family that each component of C_{ij} implies.
 - State the independence assumption required for the product form $\exp\left(-\frac{\alpha}{2} D_M^2/\chi_{\text{gate}}^2\right) \cdot \exp\left(-\frac{\beta}{2}(1 - \cos(\mathbf{e}_i, \mathbf{e}_j))\right)$ and explain what goes wrong if it is violated.
- (e) Explain why the Hungarian algorithm applied to C_{ij} enforces a global one-to-one constraint that row-wise softmax attention does not. When is the one-to-one constraint essential in multi-target tracking?

Problem 2: Product-of-Experts, Gated Fusion, and Classical Bayesian Fusion

The lecture presents three fusion operators: concatenation+MLP, weighted sum, and gated fusion. It also introduces Product-of-Experts (PoE) as a bridge to classical Bayesian fusion.

- (a) For two Gaussian experts

$$p(\mathbf{z} \mid \mathbf{y}^{(1)}) = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1), \quad p(\mathbf{z} \mid \mathbf{y}^{(2)}) = \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2),$$

derive the PoE fused distribution $p(\mathbf{z} \mid \mathbf{y}^{(1)}, \mathbf{y}^{(2)}) \propto p(\mathbf{z} \mid \mathbf{y}^{(1)})p(\mathbf{z} \mid \mathbf{y}^{(2)})$. Show that the fused precision is $\boldsymbol{\Lambda}_{\text{fused}} = \boldsymbol{\Lambda}_1 + \boldsymbol{\Lambda}_2$ and derive $\boldsymbol{\mu}_{\text{fused}}$. Compare this result explicitly with the Kalman filter measurement update when $\mathbf{y}^{(1)}$ is the prior prediction and $\mathbf{y}^{(2)}$ is the measurement likelihood.

- (b) The PoE assumes conditional independence: $p(\mathbf{y}^{(1)}, \mathbf{y}^{(2)} | \mathbf{z}) = p(\mathbf{y}^{(1)} | \mathbf{z}) p(\mathbf{y}^{(2)} | \mathbf{z})$. Give a concrete example from camera–radar fusion where this assumption is violated. What is the consequence of applying PoE when the assumption fails? (*Hint*: consider the fused covariance.)
- (c) The gated fusion operator is

$$\mathbf{g} = \sigma(\mathbf{W}_g [\mathbf{e}_{\text{cam}}; \mathbf{e}_{\text{rad}}]), \quad \mathbf{e}_{\text{fuse}} = \mathbf{g} \odot \mathbf{e}_{\text{cam}} + (1 - \mathbf{g}) \odot \mathbf{e}_{\text{rad}}.$$

Show that this is a special case of confidence-weighted fusion $\mathbf{e}_{\text{fuse}} = \sum_m w_m \mathbf{e}_m$ when the weights are element-wise and input-dependent. Then argue that gated fusion can be interpreted as a learned, feature-level analog of the classical reliability-weighted fusion rule $w_m \propto 1/\sigma_m^2$.

- (d) A system uses MoE fusion with $K = 3$ experts specialized for {clear weather, fog, night}. During training, expert collapse occurs: one expert handles 95% of samples. Propose two regularization strategies to prevent collapse and explain the mechanism of each.
- (e) Consider deploying the gated fusion system from (c) on an autonomous vehicle where the radar occasionally fails completely (outputs zeros). Explain what the gate should learn to do in this failure mode. Then explain why naïve concatenation+MLP would handle this scenario poorly without explicit data augmentation.
- (f) Suppose you want to add a third modality (LiDAR) to the gated fusion. Write the mathematical extension for three modalities using a softmax gate and explain why the two-modality sigmoid formulation does not generalize directly.

Problem 3: Attention as Data Association — Derivation and Comparison with JPDA

The lecture draws a parallel between transformer attention weights A_{ij} and JPDA association probabilities β_{ij} .

- (a) In JPDA, association probabilities are computed by marginalizing over all valid joint events:

$$\beta_{ij} = \sum_{\theta: \theta(i)=j} P(\theta | Y_{1:k}),$$

where θ ranges over all feasible joint association hypotheses. Write out the explicit computation for a scenario with 2 tracks and 3 measurements (including the possibility of missed detection for each track). How many joint events exist? Compute the marginal β_{ij} in terms of individual likelihoods.

- (b) In attention-based association:

$$\alpha_{ij} = \text{softmax}_j(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}),$$

where $\mathbf{q}_i$ is the track query and $\mathbf{k}_j$ is the detection key. Identify the precise structural differences between α_{ij} and β_{ij} in terms of row normalization, missed detections and probabilistic interpretation. Which constraints does JPDA enforce that attention does not?

- (c) A recent line of work proposes replacing softmax attention with Sinkhorn normalization to enforce approximate one-to-one matching. The Sinkhorn operator alternately normalizes rows and columns of the similarity matrix. Explain why this is a closer approximation to the optimal transport / Hungarian assignment than plain softmax. What is the computational complexity of K Sinkhorn iterations for an $N \times M$ cost matrix?
- (d) The MLP compatibility function

$$\ell_{ij} = f_{\theta}(\varphi(\mathbf{z}_i), \psi(\mathbf{y}_j))$$

replaces dot-product attention. Derive the gradient $\partial \ell_{ij} / \partial \varphi(\mathbf{z}_i)$ for a single-hidden-layer MLP with ReLU activation. Compare the expressiveness of this gradient update with the gradient of dot-product attention.

- (e) Consider a scenario with 10 tracks and 50 candidate detections (typical in dense traffic). Compare JPDA, Hungarian algorithm, and multi-head attention in terms of: (i) computational complexity, (ii) ability to represent association uncertainty, (iii) gradient flow for end-to-end training. Conclude with a recommendation for a real-time autonomous driving system.

Problem 4: System-Level Design and State-of-the-Art Comparison

- (a) You are designing a perception system for a C-UAS (Counter-Unmanned Aircraft System) that must detect and track small drones using EO/IR cameras and radar. The system must operate in real-time (< 100 ms latency) with calibrated uncertainty outputs. Design the system by specifying:
 - (i) the fusion taxonomy level (early / mid / late),
 - (ii) the association method,
 - (iii) the temporal model,
 - (iv) the uncertainty quantification approach.

Justify each choice against alternatives.

- (b) The nuScenes benchmark evaluates 3D object detection using metrics including mAP, NDS, and per-class performance. Define and describe these metrics. A BEV fusion model achieves state-of-the-art mAP but has 300 ms inference time. A late-fusion baseline achieves 8% lower mAP but runs at 50 ms. For the C-UAS system in (a), argue which system you would deploy and why. Relate your answer to the RMSE-vs-CRPS discussion from the forecasting assignment (Question 2) : why is mAP alone an insufficient metric for safety-critical deployment?
- (c) The lecture identifies conformal prediction as a research frontier for safety guarantees. Explain how split conformal prediction could be applied to BEV-based 3D detection to produce detection boxes with guaranteed marginal coverage. State the exchangeability assumption and discuss whether it is realistic in autonomous driving.

Question 2 - Paper analysis

Read one of these 2 papers and answer the following:

- **TimeGrad**: Rasul, K., Seward, C., Schuster, I., Vollgraf, R. (2021). Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In M. Meila and T. Zhang (Eds.), Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139, pp. 8857–8868). PMLR.
- **CSDI**: Tashiro, Y., Song, J., Song, Y., Ermon, S. (2021). CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In Advances in Neural Information Processing Systems (Vol. 34, pp. 24804–24816). NeurIPS.

To address the following problems, always select one paper and stick with the same paper.

Problem 1: Why conditional diffusion is the correct probabilistic object

Choose **either TimeGrad or CSDI**. Let y denote the unknown quantity of interest: for TimeGrad, $y = x_{t_0:T}$ (future trajectory), and for CSDI, $y = x_{\text{mis}}$ (missing values). Let c denote the available context: for TimeGrad, $c = (x_{1:t_0-1}, \text{covariates})$; for CSDI, $c = (x_{\text{obs}}, m)$ where m is the observation mask.

- State precisely whether the paper learns $p_\theta(y)$ or $p_\theta(y | c)$.
- Suppose a student proposes to train an *unconditional* diffusion model on full trajectories and then, at test time, simply keep the generated sample entries that happen to match the observed context “approximately.” Give two rigorous reasons why this is not the same as learning the correct conditional distribution.
- For your chosen paper, explain what random variable is diffused and what variable is *not* diffused. Why is this distinction important?
- Relate this to the course distinction between

point prediction, conditional density estimation, posterior sampling.

Which of these is the paper really performing?

Problem 2: Architectural decomposition and the logic of the conditioning pathway

Choose **TimeGrad or CSDI** and answer the following.

- Write a block-level decomposition of the model in the form
context encoder $\rightarrow$ conditioning representation $\rightarrow$ denoiser $\rightarrow$ reverse diffusion sampler.
Describe the role of each block.
- What information enters the denoiser *at every reverse step*? Why is conditioning only once at the beginning generally weaker?
- TimeGrad uses an RNN-based context pathway, whereas CSDI uses observation masks plus temporal/feature attention. Explain why these are not arbitrary choices, but are matched to the probabilistic structure of the task.
- In your course, you distinguish between

sequence modeling, set/graph-like dependence, masked inference.

Explain where your chosen model sits in that taxonomy.

Problem 3: Uncertainty quantification — what is represented, and what is not?

Consider either paper.

- (a) Is uncertainty represented explicitly by a variance parameter in the final output, or implicitly through sampling? Be precise.
- (b) Explain how you would obtain:
 - (i) a point estimate,
 - (ii) a 90% interval,
 - (iii) a calibration diagnosticfrom the model outputs.
- (c) Distinguish the following sources of uncertainty in the context of these papers:
aleatoric uncertainty, epistemic uncertainty, Monte Carlo sampling variability.

Which one is most directly captured?
- (d) A student says: “Because the model outputs 100 samples, it automatically gives calibrated uncertainty.” Critique this statement carefully.

Problem 4: Proper evaluation and why CRPS appears in both papers

- (a) Explain why RMSE or MAE alone is insufficient for evaluating these models.
- (b) Define CRPS and explain what CRPS rewards and penalizes. Explain its relationship with MAE and Brier metrics.
- (c) Why is CRPS especially suitable when the model produces samples rather than an analytic density?
- (d) For TimeGrad, explain why CRPS_{sum} is not redundant with per-dimension CRPS. What multivariate phenomenon is it trying to probe?
- (e) Propose one additional engineering evaluation beyond CRPS that you would include for
 - (i) forecasting,
 - (ii) imputation,and justify it.